Feature Processing

AI in Music, Winter Term 2025/2026

13.11.2025

Exercise 1 [Feature Extraction, 2 Points]

Load the file `rock.mp3` from the week 2 exercises in Python and extract the following features with librosa, once with a frame length of 512 and once with 760:

- Spectral centroid
- Zero-crossing rate
- Root mean squared energy (RMS)

See https://librosa.org/doc/latest/feature.html for documentation on how to extract the features.

Exercise 2 [Harmonisation of Feature Matrix, 5 Points]

Given the features from Exercise 1, calculate a harmonised feature matrix for all six feature vectors. The features should be harmonised to the frame length of $f_{min} = 512$.
First, use the harmonisation method shown on slide 18 and 19 of the lecture "Feature Processing and Selection".
Then, do the same by using a weighted average method in which features of adjacent frames are weighted by their overlap ratio with the new frame.

Exercise 3 [Correlation, 3 Points]

For each of three feature types, calculate the correlation between the features with frame length 512 from Exercise 1 and the harmonised features with original frame length 760.
Which harmonisation method achieves higher correlation with the extracted features?